

Musterlösung Serie 13

INVERTIBILITY

1. Inverses. For each of the following matrices, determine whether or not it is invertible, and if it is compute its inverse.

(a) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$

(d) $\begin{pmatrix} 1 & 2 & -3 & 1 \\ -1 & 3 & -3 & -2 \\ 2 & 0 & 1 & 5 \\ 3 & 1 & -2 & 6 \end{pmatrix}$

Lösung:

- (a) This matrix is of rank 1 since the first row is equal to the second and to the third. Therefore it cannot be transformed into the identity using elementary row-operations and it isn't invertible.

(b) $\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 4/5 & -1/5 & -1/5 & -1/5 \\ -1/5 & 4/5 & -1/5 & -1/5 \\ -1/5 & -1/5 & 4/5 & -1/5 \\ -1/5 & -1/5 & -1/5 & 4/5 \end{pmatrix}$

$$(d) \begin{pmatrix} 35/11 & -16/11 & 13/11 & -2 \\ 1 & 0 & 1 & -1 \\ 10/11 & -3/11 & 10/11 & -1 \\ -16/11 & 7/11 & -5/11 & 1 \end{pmatrix}.$$

2. Invertible square matrix. Show that a square matrix A is invertible if and only if its transpose A^T is invertible. Moreover, show that in that case we have $(A^T)^{-1} = (A^{-1})^T$.

Lösung: We already know for all quadratic matrices A and B of the same size, the equation $(A \cdot B)^T = B^T \cdot A^T$ is satisfied.

First, assume that A is invertible with inverse A^{-1} . Then:

$$A^T \cdot (A^{-1})^T = (A^{-1} \cdot A)^T = I^T = I,$$

$$(A^{-1})^T \cdot A^T = (A \cdot A^{-1})^T = I^T = I.$$

Therefore, the matrix A^T is invertible with inverse $(A^{-1})^T$ and we have $(A^T)^{-1} = (A^{-1})^T$. Second, assume A^T is invertible. We proved in the lecture $(A^T)^T = A$. Going through the first case with A^T instead of A yields that $(A^T)^T = A$ is invertible.

3. Invertible matrices.

- (a) Let V, W be two n -dimensional vector spaces over a field K . Let $S \in \text{Hom}(V, W)$ and $T \in \text{Hom}(W, V)$ such that $T \circ S = \text{Id}_V$. Show that S is invertible and T is the inverse of S .
- (b) Let $A, B \in M_{n \times n}(K)$ such that $A \cdot B = I_n$. Show that A is invertible with inverse B .

Lösung:

- (a) Since S admits a left-sided inverse, it is injective. Since W is finite-dimensional, this implies that S is surjective. Since T admits a right-sided inverse, it is surjective. So, it is injective as V is finite-dimensional. This shows that both S and T are isomorphisms, and therefore that they are invertible. Note that for any $w \in W$, there exists a unique $v \in V$ such that $S(v) = w$. Hence

$$S \circ T(w) = S \circ T(S(v)) = S(v) = w.$$

So,

$$T \circ S = \text{Id}_V \quad \text{and} \quad S \circ T = \text{Id}_W,$$

which implies that T is the inverse of S by definition.

- (b) A , respectively B , defines a linear map $T_A : K_{\text{col}}^n \rightarrow K_{\text{col}}^n$, $v \mapsto A \cdot v$, respectively $T_B : K_{\text{col}}^n \rightarrow K_{\text{col}}^n$, $v \mapsto B \cdot v$. Since $A \cdot B = I_n$, for any $w \in K_{\text{col}}^n$ we have

$$T_A \circ T_B(w) = T_A(B \cdot w) = A \cdot (B \cdot w) = (A \cdot B) \cdot w = w.$$

So, $T_A \circ T_B = \text{Id}_{K_{\text{col}}^n}$ and by (a), we obtain that T_A and T_B are invertible and are the inverse of each other. Equivalently

$$B \cdot A = A \cdot B = I_n,$$

so A and B are invertible with $B = A^{-1}$.

4. Invertible similar matrices. Consider $n \times n$ -matrices A and B over K .

- (a) Show: If A or B is invertible, then AB and BA are similar.
 (b) Does this also hold true without the condition in (a)?

Lösung:

- (a) If A is invertible, then $AB = A(BA)A^{-1}$ is similar to BA . Likewise, if B is invertible, then $BA = B(AB)B^{-1}$ is similar to AB .
 (b) For $A := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B := \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ we have $AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $BA = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. As the zeromatrix is only similar to itself, this is a counterexample.

5. Gaussian elimination and invertibility. Determine with the help of Gaussian elimination, for which values of α the following matrix over $\mathbb{Q}$ is invertible:

$$\begin{pmatrix} 1 & 3 & -4 & 2 \\ 2 & 1 & -2 & 1 \\ 3 & -1 & -2 & -2\alpha \\ -6 & 2 & 1 & \alpha^2 \end{pmatrix}.$$

Lösung: Gaussian elimination yields the matrix

$$\begin{pmatrix} 1 & 3 & -4 & 2 \\ 0 & -5 & 6 & -3 \\ 0 & 0 & -2 & -2\alpha \\ 0 & 0 & 0 & \alpha^2 - \alpha \end{pmatrix},$$

which is invertible if and only if the original matrix is invertible. As a triangular matrix is invertible if and only if it has non-zero diagonal entries, it follows that the original matrix is invertible if and only if

$$\alpha^2 - \alpha = \alpha(\alpha - 1) \neq 0.$$

This is equivalent to $\alpha \notin \{0, 1\}$.

6. Bruhat decomposition. Prove the following:

Theorem. (*Bruhat-Decomposition*) For every invertible matrix A there exists a permutation matrix P , i.e. a matrix which has exactly one non-zero, which equals 1, in every column and every row, and invertible upper triangular matrices B and B' , such that

$$A = BPB'.$$

Hint: Choose an invertible upper triangular matrix U , such that the sum of the number of leading zeros in all rows of UA is maximal. Then find a permutation matrix Q , such that QUA is an invertible upper triangular matrix.

Multiplying a matrix A on the left by a permutation matrix P permutes the rows of A . For example, if P has a 1 at index (i, j) , the j -th line of A will become the i -th line of PA .

Lösung: For every invertible upper triangular matrix U we consider the sum of the number of leading zeros of all rows of UA . This sum is bounded by the number of entries of A , and hence we can choose an invertible upper triangular matrix U which maximises this sum.

Claim. The numbers of leading zeros of the rows of UA are pairwise different.

Proof. Assume that the claim is false for the i -th and j -th row with $i < j$. Then the first non-zero entry of these rows is in the same column. We subtract a suitable multiple of the j -th row from the i -th row. The i -th row of the resulting matrix has at least one leading zero more. This row operation corresponds to left multiplication with an invertible upper triangular matrix of the Form $T = I_m + \lambda E_{ij}$. As the other rows are unchanged, the sum of the leading zeros of TUA is greater than the one of UA . As TU also is an invertible upper triangular matrix, this is a contradiction to the maximised choice of U . $\square$

The claim yields that we can transform UA into an upper triangular matrix B' by left multiplication with a permutation matrix Q . In other words, we have $QUA = B'$. As Q , U and A are invertible, so is B' . The inverse of a permutation matrix is again a permutation matrix and the inverse of an upper triangular matrix is again an upper triangular matrix. Hence with $P := Q^{-1}$ and $B := U^{-1}$ we get the equation $A = BPB'$ of the desired form.

Single Choice. In each exercise, exactly one answer is correct.

1. Which assertion is not always satisfied?
 - (a) The base change matrix is the representation matrix of the identity map with respect to the according bases.
 - (b) Every finite dimensional vector space is isomorphic to K^n for some $n \geq 0$.
 - ✓(c) The rank of a linear map $f : K^n \rightarrow K^m$ is at least $\min\{n, m\}$.
 - (d) The representation matrix of an isomorphism is invertible.

Erklärung: The rank of a linear map $f : K^n \rightarrow K^m$ is at *most* $\min\{n, m\}$; but can for example also be 0, thus (c) is false.

2. Consider $\mathbb{C}$ as real two-dimensional vectorspace with the ordered basis $\mathcal{B} := (1, i)$. The matrix $[\dots]_{\mathcal{B}}^{\mathcal{B}} := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is the representation matrix with respect to $\mathcal{B}$ of the linear map $\mathbb{C} \rightarrow \mathbb{C}$:

- (a) Complex conjugation $z \mapsto \bar{z}$
- (b) $z \mapsto \operatorname{Re}(z)$
- (c) $z \mapsto \operatorname{Im}(z)$
- ✓(d) $z \mapsto iz$

Erklärung: In the Basis $\mathcal{B}$ an element $a + ib \in \mathbb{C}$ is written as the vector $\begin{pmatrix} a \\ b \end{pmatrix}$. Under the matrix $[\dots]_{\mathcal{B}}^{\mathcal{B}}$, it is mapped to $\begin{pmatrix} -b \\ a \end{pmatrix}$, which corresponds to the element $-b + ia = i(a + ib)$.

3. The rank of $\begin{pmatrix} 1 & 3 \\ -3 & -9 \end{pmatrix}$ over $\mathbb{Q}$ is

- (a) 0
- ✓(b) 1
- (c) 2
- (d) 3

Erklärung: The columns of the matrix are non-zero, but linearly dependent. The second column is -3 times the first one. Thus, the rank is 1.

4. For every $n \times m$ -Matrix A and every invertible $n \times n$ -matrix B , we have

- (a) $\text{rank}(BA) = \text{rank}(B) \cdot \text{rank}(A)$
- (b) $\text{rank}(BA) = \text{rank}(B) + \text{rank}(A)$
- ✓(c) $\text{rank}(BA) = \text{rank}(A)$
- (d) $\text{rank}(BA) = \text{rank}(B)$

Erklärung: According to Lemma 3.4.2 of the lecture, the rank of a linear map does not change by left composition with an isomorphism. In the language of matrices this translates to (c).

Multiple Choice Fragen.

1. Consider the following ordered bases of $\mathbb{R}[x]_2$:

$$\mathcal{B} = (1, x, x^2), \quad \mathcal{C} = (x^2, (x+1)^2, (x+2)^2)$$

Determine the base change matrix $Q := [\text{Id}_{\mathbb{R}[x]_2}]_{\mathcal{C}}^{\mathcal{B}}$.

(a) $Q = \begin{pmatrix} 0 & 1 & 4 \\ 0 & 2 & 4 \\ 1 & 1 & 1 \end{pmatrix}$

(b) $Q = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 1 & 4 \end{pmatrix}$

(c) $Q = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 2 & 1 \\ 4 & 4 & 1 \end{pmatrix}$

(d) $Q = \frac{1}{4} \begin{pmatrix} 2 & -4 & 4 \\ -1 & 4 & -3 \\ 4 & 0 & 0 \end{pmatrix}$

✓ (e) $Q = \frac{1}{4} \begin{pmatrix} 2 & -3 & 4 \\ -4 & 4 & 0 \\ 2 & -1 & 0 \end{pmatrix}$

(f) $Q = \frac{1}{4} \begin{pmatrix} 2 & -1 & 4 \\ -4 & 4 & 0 \\ 4 & -3 & 0 \end{pmatrix}$

Erklärung: One solves the system of equations

$$p = ax^2 + b(x+1)^2 + c(x+2)^2 = (a+b+c)x^2 + (2b+4c)x + (b+4c)$$

for $p = 1, x, x^2$ using comparison of coefficients. For $p = 1$, we get

$$b + 4c = 1$$

$$2b + 4c = 0$$

$$a + b + c = 0$$

The second equation yields $c = -\frac{1}{2}b$ and substitution into the first one $b = -1$. Hence we get $c = \frac{1}{2}$. Substitution into the third equation implies and thus

$$[1]_{\mathcal{C}} = \begin{pmatrix} \frac{1}{2} \\ -1 \\ \frac{1}{2} \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$$

This already excludes all but one answers. Of course one could now also compute $[x]_{\mathcal{C}}$, and $[x^2]_{\mathcal{C}}$.

2. Let $A \in M_{n \times n}(K)$ and $P \in \text{GL}_n(K)$, denote

$$r := \text{rank}(A - I_n) \quad \text{and} \quad s := \text{rank}(PAP^{-1} - I_n).$$

Which of the following statements hold?

✓ (a) $r = s$; *Erklärung*: Note that

$$P(A - I_n)P^{-1} = PAP^{-1} - PI_nP^{-1} = PAP^{-1} - I_n.$$

Hence $PAP^{-1} - I_n$ and $A - I_n$ are equivalent matrices. As seen in the lectures, this implies $r = s$.

(b) $r \neq s$;

(c) $r > s$;

(d) $r < s$;

✓ (e) if $r < n$, $\exists v \in V \setminus \{0\}$ such that $Av = v$.

Erklärung: As usual, $A - I_n$ defines a linear map

$$\begin{array}{ccc} T : K_{\text{col}}^n & \rightarrow & K_{\text{col}}^n \\ v & \mapsto & (A - I_n) \cdot v \end{array}$$

So,

$$n = \dim(K_{\text{col}}^n) = \dim(\ker(T)) + \text{rank}(T) = \dim(\ker(T)) + r.$$

If $r < n$, $\dim(\ker(T)) \geq 1$ so there exists some $v \in \ker(T) \setminus \{0\}$. Equivalently, there exists some non-zero $v \in V$ such that

$$A \cdot v - I_n \cdot v = 0 \Leftrightarrow A \cdot v = v.$$